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Blockchain-based Traceability Solutions in Fisheries: Insights from Greece

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RESEARCH QUESTION: What is the Greek farmers' intention to adopt a blockchain-based traceability solution?

Introduction

Blockchain technology provides several benefits when applied to circular economy, enhancing transparency, traceability, resource management and collaboration. It can help businesses and consumers transition to more sustainable, circular practices that focus on reducing waste and conserving resources. On the other hand, global fisheries and aquaculture production has steadily expanded in recent years. The fisheries supply chain is a complicated network involving multiple stakeholders. There is a need for more efficient, traceable, and transparent fisheries supply chains. Efficient supply chain management can optimize resource use and minimize waste, contributing to the overall health of marine ecosystems towards sustainability.

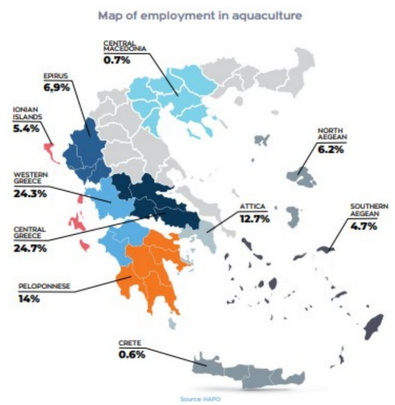
Review

Typical examples of blockchain implementations in fisheries:

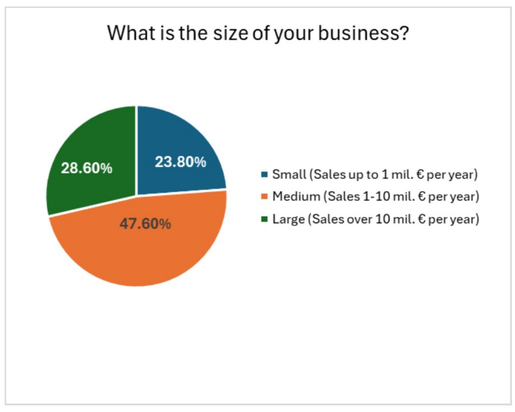
- **FishCoin:** It uses blockchain to enable traceability in the seafood industry and provides transparency, giving consumers and buyers greater insight into the product's journey and quality.
- **Bumble Bee Seafoods:** It uses blockchain technology to enable full traceability of its seafood products, giving consumers a transparent view of the seafood's journey from catch to plate. **Provenance:** It supports blockchain technology to trace the journey of tuna from fishermen in Southeast Asia to supermarket shelves.
- **Seafood Traceability:** IBM Food Trust supports a technology that allows for the end-to-end tracking of seafood products, ensuring that they are sustainably sourced and meet regulatory requirements.

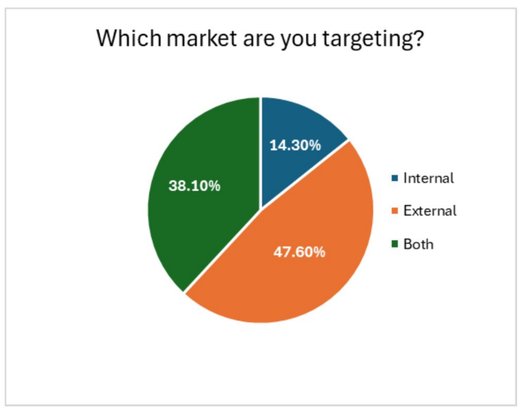
Methodology

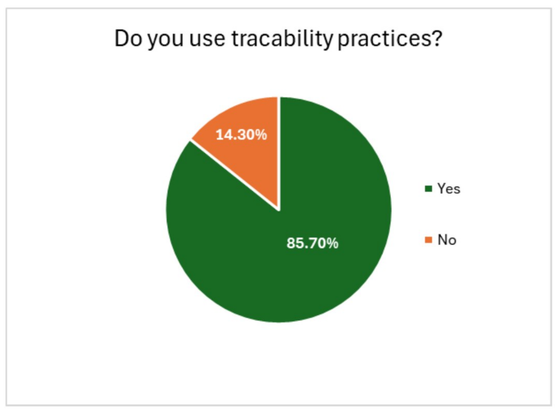
- Online survey using google forms
- Structured questionnaire based on TAM model
- Participants: 21 Greek fish farmers
- Survey period: June 2024

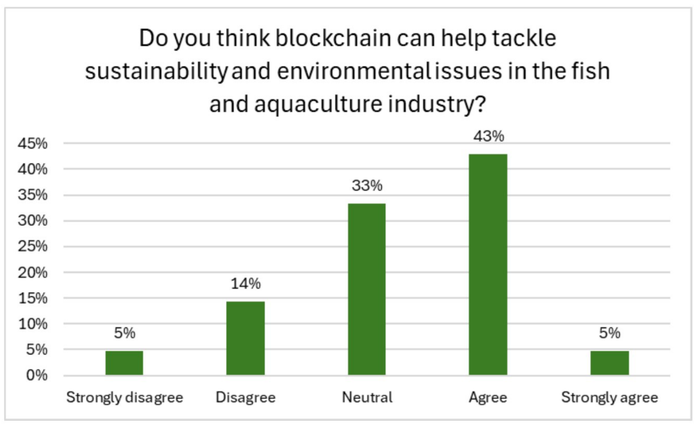


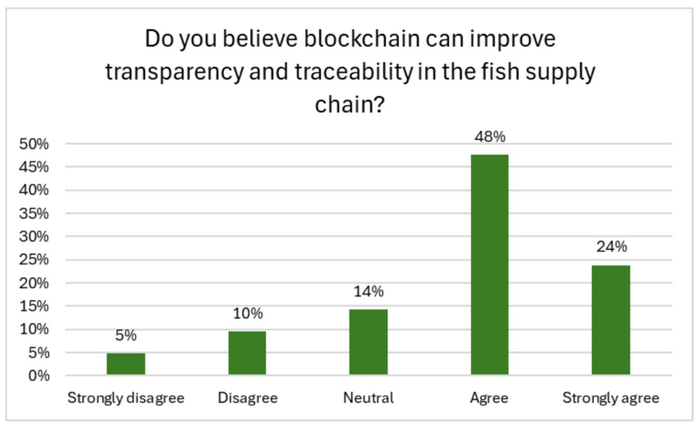
Results

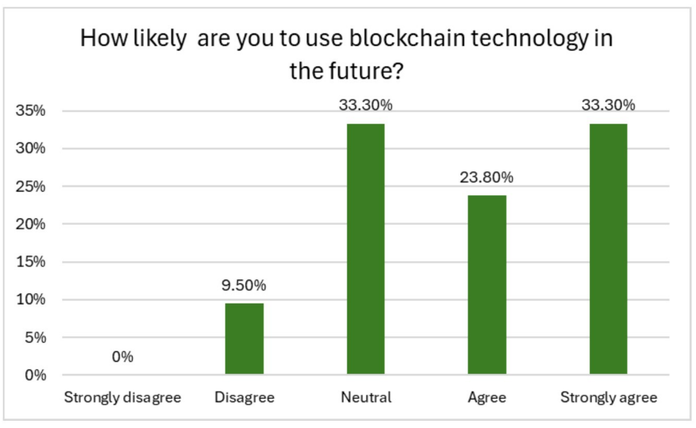












Conclusions

Sustainable fisheries supply chains play a crucial role in preserving marine biodiversity and supporting the transition to a circular economy. Blockchain traceability solutions can enhance transparency, allowing for more efficient resource management and minimizing waste, which aligns with circular economy principles. According to survey results, Greek fish farmers are open to adopting blockchain traceability but remain thoughtful about its perceived benefits. The participants indicate that the main barriers are the complexity of blockchain technology, which requires technical expertise; initial setup and maintenance costs; and lack of information and training from public authorities. Despite these challenges, the evolution of blockchain technology offers significant potential to drive sustainability in fisheries by improving traceability, reducing resource waste, and facilitating a circular economy model.

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